Lecture Scribe Creation

Lecture_Content.pdf
PDF

Structure_Example_Scribe...
PDF

I'm preparing lecture notes for exam revision and want them to follow a consistent structure throughout the semester for all my lecture scribe.

For that I am sharing you two pdf files.
--> The first PDF file consists of the lecture PDF for which you need to make a complete lecture scribe.
--> The second PDF file consists of the example of structure (template) of the lecture scribe, which is provided only to demonstrate the structure of how the scribe must look like it's formatting style and all.
Do not copy or paraphrase any content from the example scribe.
NOTE: The content of this example file must not be included in the final scribe (only refer to the structure). It is only given to know that how to scribe must be organized.

The goal of this scribe is to help with exam revision.
The scribe must cover these points properly:
- Cover the entire lecture PDF.
- It should be written as exam revision material style.
- Include all the definitions, formulas, equations, derivations, proofs, and all the examples only if they are in the lecture PDF, don't introduce new content.
- Please stick only to the content which appears in the lecture PDF.
- And give derivations for only those which are given, don't include extra or new things.
- And also don't omit any concept from the PDF.
- Maintain a proper academic note taking style similar to well-written human lecture notes.
- Also maintain the order of the topics as presented in the lecture PDF.

How the output should be:
- Follow the structure example file for template of the scribe and do not include any content of it.
- Do not include any content from the example scribe.
- Also please do not include anything in the output which includes any reasoning or explanation of how the scribe was created.

Start generating the lecture scribe only after carefully analyzing both the files.

Lecture 20: Linear Transformations and Expectations of Multiple Random Variables

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1. Overview of the Lecture

This lecture focuses on the statistical representation of multiple random variables using vector form and studies how their statistical properties behave under linear transformations.

Main topics covered:

- Random vector representation
- Mean vector, correlation matrix, covariance matrix
- Linear transformations of random vectors
- Transformation of mean, correlation, and covariance
- Gaussian distributions (1D, 2D, and multivariate)
- Properties of uncorrelated Gaussian variables
- Application through a smart city case study

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2. Key Definitions and Terminology

Random Vector

A collection of random variables represented in vector form:

$$X = [X_1, X_2, \ldots, X_N]^T$$

- Provides a compact representation of multiple random variables

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Mean Vector

$$\mu_X = E[X]$$

- Represents the average value of the random vector
- Each element corresponds to:

$$\mu_X = [E[X_1], E[X_2], \ldots, E[X_N]]^T$$

Correlation Matrix

$$R_{XX} = E[XX^T]$$

- Element-wise:

$$R_{XX}(i,j) = E[X_i X_j]$$

- Properties:
  - Symmetric matrix
  - Size: $N \times N$

Covariance Matrix

$$C_{XX} = E[(X - \mu_X)(X - \mu_X)^T]$$

- Represents dependencies between variables
- Structure:

$$C_{XX} = \begin{bmatrix} \mathrm{Var}(X_1) & \mathrm{Cov}(X_1, X_2) & \cdots \\ \mathrm{Cov}(X_2, X_1) & \mathrm{Var}(X_2) & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix}$$

- Symmetric:

$$\mathrm{Cov}(X_i, X_j) = \mathrm{Cov}(X_j, X_i)$$

Linear Transformation

$$Y = AX + b$$

Where:

- A: Transformation matrix
- b: Constant vector
- X: Input random vector

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3. Explanation of Core Concepts and its Understanding

Vector Representation

Multiple random variables are treated as a single entity (vector), enabling matrix-based operations.

Statistical Relationships

- Mean → central tendency
- Correlation → raw dependency
- Covariance → dependency relative to mean

Effect of Linear Transformation

Given:

$$Y = AX + b$$

- Scaling/rotation → via A
- Shifting → via b

Gaussian Framework

- Used to model multivariate systems
- Defined by:
  - Mean vector
  - Covariance matrix

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4. Important Formulas and Mathematical Expressions

Mean

$$\mu_X = E[X]$$

Correlation Matrix

$$R_{XX} = E[XX^T]$$

Covariance Matrix

$$C_{XX} = E[(X - \mu_X)(X - \mu_X)^T]$$

Linear Transformation

$$Y = AX + b$$

Mean of Transformed Vector

$$\mu_Y = E[Y] = A\mu_X + b$$

Correlation Transformation

$$R_{YY} = AR_{XX}A^T + A\mu_X b^T + b\mu_X^T A^T + bb^T$$

Covariance Transformation

$$C_{YY} = AC_{XX}A^T$$

Univariate Gaussian

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

Joint Gaussian (2D)

$$f_{XY}(x,y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp(\cdots)$$

Multivariate Gaussian

$$f_X(x) = \frac{1}{\sqrt{(2\pi)^N \det(C_{XX})}} \exp\left(-\frac{1}{2}(x-\mu_X)^T C_{XX}^{-1}(x-\mu_X)\right)$$

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5. Derivations / Proofs

Mean Transformation

Step 1: Start

$$\mu_Y = E[Y]$$

Step 2: Substitute

$$= E[AX + b]$$

Step 3: Apply linearity

$$= E[AX] + E[b]$$

Step 4: Simplify

$$= AE[X] + b$$

Final Result

$$\mu_Y = A\mu_X + b$$

Correlation Transformation

Start

$$R_{YY} = E[YY^T]$$

Substitute

$$= E[(AX + b)(AX + b)^T]$$

Expand

$$= E[AXX^TA^T + AXb^T + bX^TA^T + bb^T]$$

Apply Expectation

$$= AR_{XX}A^T + A\mu_X b^T + b\mu_X^T A^T + bb^T$$

Covariance Transformation

Key Observation

$$Y - \mu_Y = A(X - \mu_X)$$

Start

$$C_{YY} = E[(Y - \mu_Y)(Y - \mu_Y)^T]$$

Substitute

$$= E[A(X - \mu_X)(X - \mu_X)^T A^T]$$

Final Result

$$C_{YY} = AC_{XX}A^T$$

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6. Worked Examples / Applications

Example: Numerical Linear Transformation

Given:

- $\mu_X = [2, -1, 1]^T$
- A matrix
- R_{XX} matrix

(a) Mean

$$\mu_Y = A\mu_X$$

Result

$$\mu_Y = [1, 2, -3]^T$$

(b) Correlation Matrix

$$R_{YY} = AR_{XX}A^T$$

Steps:

1. Compute AR_{XX}
2. Multiply with A^T

Result

$$R_{YY} = \begin{bmatrix} 17 & -6 & -11 \\ -6 & 14 & -8 \\ -11 & -8 & 19 \end{bmatrix}$$

(c) Covariance Matrix

$$C_{YY} = R_{YY} - \mu_Y \mu_Y^T$$

Result

$$C_{YY} = \begin{bmatrix} 16 & -8 & -8 \\ -8 & 10 & -2 \\ -8 & -2 & 10 \end{bmatrix}$$

Homework Example (Mean Extraction)

$$\mu_X \mu_X^T = R_{XX} - C_{XX}$$

- Mean values obtained from diagonal elements

Case Study: Smart City System (Page 55–60)

- Random vector:

$$X = [X_1, X_2, X_3, X_4, X_5]^T$$

Where:

- Traffic flow
- Speed
- Rainfall
- Temperature
- Pollution
- Covariance interpretation:
  - Rain ↑ → Traffic ↑
  - Rain ↑ → Speed ↓
  - Traffic ↑ → Pollution ↑
- Transformation:

$$Y = 0.6X_1 - 0.4X_2 + 0.8X_3$$

- Result:
  - Y is Gaussian
  - Mean:

$$E[Y] = a^T \mu$$

- Variance:

$$\mathrm{Var}(Y) = a^T \Sigma a$$

7. Important Observations and Results

- Correlation matrix is symmetric
- Covariance matrix represents dependency structure
- Linear transformation:
  - Mean depends on both A and b
  - Covariance depends only on A
- Shift b does NOT affect covariance
- Gaussian distributions are fully defined by mean and covariance
- Uncorrelated Gaussian variables ⇒ covariance matrix is diagonal
- Joint Gaussian factorizes into product of individual PDFs when uncorrelated

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8. Quick Revision Points

- $X = [X_1, ..., X_N]^T$
- $\mu_X = E[X]$
- $R_{XX} = E[XX^T]$
- $C_{XX} = E[(X - \mu_X)(X - \mu_X)^T]$
- Linear transformation:
  - $Y = AX + b$
  - $\mu_Y = A\mu_X + b$
  - $C_{YY} = AC_{XX}A^T$
- Gaussian:
  - Defined by μ and C
  - Multivariate PDF important
- Key insight:
  - Covariance = “dependency map”
  - Mean = “central behavior”

End of Lecture Scribe